Certification Evidence Matrix For Public Election Count Audits

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Abstract

Election certification is a legal and administrative act, not a single hash or software verdict. RCOUNT therefore needs an evidence matrix: a way to say which public records support each mechanical claim, which verifier equation checks that claim, and which certification questions remain outside the package. This paper maps recurring certification evidence families to implemented RCOUNT fixtures: canvass arithmetic, precinct lineage, batch accounting, privacy-safe inclusion, CVR reconciliation, RLA replay and stopping checks, jurisdiction adapter metadata, district aggregation, and manual audit reconciliation. The matrix is designed to help officials, researchers, and courts distinguish machine-checkable public evidence from statutory judgment.

1 Introduction

Certification compresses many different claims into one public moment. Some claims are arithmetic: do precinct totals sum to jurisdiction totals? Some are custodial: do ballot manifests and batches account for the counted ballots? Some are procedural: did the required audit, recount, or canvass step occur under the right authority? The EAC frames canvass and certification as the path from unofficial to official results, supported by checklists, communication, and post-election procedures [U.S. Election Assistance Commission, 2026a].

RCOUNT does not replace that process. It gives the mechanical parts a stable public transcript. A verifier can recompute totals, hashes, audit samples, and hand-count comparisons. It cannot decide voter intent, cure a ballot, accept a late return, certify equipment, or substitute for a statutory officer.

The purpose of this paper is to make that boundary visible. The evidence matrix below connects each certification evidence family to RCOUNT records, verifier equations, passing fixtures, failing fixtures, and remaining human decisions.

2 Evidence Families

Across states, certification evidence tends to fall into recurring families. The exact deadline, official, and audit type varies, but the public proof problem is stable enough to model.

Evidence family	Public question
Canvass arithmetic	Do reporting-unit, contest, batch, and jurisdiction totals agree after official corrections?
Lineage	Are precinct splits, merges, renames, and boundary changes explicit across cycles?
Source integrity	Do public exports, summaries, CVRs, and manifests still hash to the declared package evidence?
Ballot accounting	Do batches, manifests, accepted ballots, rejected ballots, and counted ballots reconcile?
CVR reconciliation	Do ballot-level CVR rows reproduce published summary totals where CVRs are available?
Audit completion	Did the required hand audit, RLA, or recount produce an auditable pass/escalate record?
Voter-facing inclusion	Can a voter check inclusion without creating a receipt for candidate choices?
District aggregation	Do district totals follow from a count package and a plan/context hash?

Risk-limiting audits deserve special care because they are statistical and jurisdiction-specific. NCSL describes RLAs as audits that limit the risk of certifying an incorrect outcome and notes that audit completion may be tied to canvass or certification timelines [National Conference of State Legislatures, 2026]. RCOUNT therefore records replayable samples, observations, margins, stopping methods, and adapter metadata, but leaves statutory adequacy to jurisdiction-specific rules.

3 Implemented Matrix

RCOUNT check	Evidence used	What it mechanically proves
`contest_selection_sum`	Contest summaries	Selection totals plus residual counts equal counted ballots.
`jurisdiction_contest_total`	Reporting-unit and jurisdiction summaries	Local totals roll up to the jurisdiction total.
`batch_summary_total`	Batch manifests and batch-tagged summaries	Counted, accepted, and rejected ballot accounting is internally consistent.
`lineage_conservation`	Prior/current reporting units and lineage records	Split/merge records reference real units and have valid cardinality.
`source_hash_match`	Source index and package-relative files	Published source bytes match the declared hashes.
`proof_privacy_gate`	Inclusion proof records	Voter-facing proofs do not disclose candidate selections or linkable voter metadata.
`cvr_summary_total`	CVR rows and public summaries	CVR aggregates reproduce public reporting-unit summaries.
`rla_sampler_replay`	CVR population, seed, manifest hash, and algorithm id	Published RLA sample draws replay from public material.
`rla_margin_metadata`	RLA margin record and canvassed summary	Reported audit margin fields match public totals.
`rla_stopping_rule`	Sample observations, discrepancies, margin, and risk estimate	Declared pass/escalate status follows from the named RCOUNT stopping contract.
`rla_jurisdiction_adapter`	Jurisdiction method metadata	State-style record-shape requirements are present for synthetic Colorado and California fixtures.
`manual_audit_reconciliation`	Manual audit hand totals and canvassed machine totals	Hand-count totals are within declared tolerance, and declared status agrees.

This matrix is intentionally operational. It names equation ids visible in `rcount verify` transcripts, not abstract desiderata. A public reader can ask which row failed, which evidence file fed that row, and whether the failure is arithmetic, hashing, privacy, audit, or adapter metadata.

4 Model Fixtures

The current fixture suite contains three model audit styles.

Fixture	Positive claim	Negative pair
`colorado-rla`	A Colorado-style comparison audit record has a 20 digit public seed, SHA-256 sampler, margin metadata, stopping method, risk estimate, and status.	`bad-colorado-rla` rejects a seed that is not 20 decimal digits.
`california-rla`	A California-style public audit record names ballot manifest format, audit software id, public source URL, margin, and status.	`bad-california-rla` rejects non-public source metadata.
`manual-audit`	A plain manual audit hand count matches canvassed machine totals within zero-vote tolerance.	`bad-manual-audit` catches a hand-count drift hidden behind declared pass.

These fixtures are not real state certifications. They are controlled examples that exercise different evidence styles. Colorado's public RLA materials describe ballot manifests, public seed generation, and audit software sampling [Colorado Secretary of State, 2026a]; Rule 25 also names CVR and summary reconciliation requirements for comparison audits [Colorado Secretary of State, 2026b]. California's RLA regulations emphasize an accurate ballot manifest and public software algorithm/source-code disclosure [California Secretary of State, 2026]. The manual audit fixture covers the more ordinary case: compare hand-count totals to the canvassed machine totals and escalate when they disagree beyond tolerance.

5 Boundaries

The evidence matrix is not a certification rulebook. It is a public-verifier map. A passing RCOUNT package can show that particular records agree, that declared hashes match the bytes, that audit samples replay, and that hand-count totals fall within a stated tolerance. It does not prove that every eligible voter was registered, every ballot was correctly interpreted, every chain of custody question was resolved, or every statutory deadline was met.

Voting-system certification is also separate. The EAC voting system program tests and certifies voting equipment; RCOUNT checks exported public evidence after the fact and does not certify the equipment or vendor software [U.S. Election Assistance Commission, 2026b].

The matrix should therefore be read as a set of labels for mechanical evidence. Those labels can support certification work, recounts, litigation, public records review, and independent research, but they do not move legal authority from officials to software.

6 Conclusion

RCOUNT now has enough fixture breadth to support a certification evidence matrix. It can model ordinary canvass arithmetic, ballot and batch accounting, precinct lineage, privacy-safe inclusion, CVR reconciliation, replayable RLAs, state-style RLA metadata, district aggregation, and a plain manual audit.

The next step is depth. The matrix should grow from synthetic examples into jurisdiction-qualified adapters: real export schemas, real audit report shapes, real recount triggers, and legally specified RLA stopping methods. The fixed point remains the same: public evidence should be recomputable, failures should name the exact broken equation, and legal certification should remain a human and statutory act.

References

California Secretary of State. Risk-limiting audits regulations. https://www.sos.ca.gov/administration/regulations/current-regulations/elections/risk-limiting-audits, 2026. Accessed 2026-05-13.

Colorado Secretary of State. Risk-limiting audit faqs. https://www.coloradosos.gov/pubs/elections/RLA/faqs.html, 2026a. Accessed 2026-05-13.

Colorado Secretary of State. Colorado election rules, rule 25. https://www.law.cornell.edu/regulations/colorado/8-CCR-1505-1-25, 2026b. Accessed 2026-05-13.

Gio Della-Libera. Rcount substrate specification. Technical report, Apportionment Research, 2026.

National Conference of State Legislatures. Post-election audits. https://www.ncsl.org/elections-and-campaigns/post-election-audits, 2026. Accessed 2026-05-13.

U.S. Election Assistance Commission. Election results, canvass, and certification. https://www.eac.gov/election-officials/election-results-canvass-and-certification, 2026a. Accessed 2026-05-13.

U.S. Election Assistance Commission. Voting equipment: System certification process. https://www.eac.gov/voting-equipment/system-certification-process, 2026b. Accessed 2026-05-13.